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The Economics of Promotion and Relegation in Sports Leagues

The Case of English Football

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In most of the world's professional sports leagues, the worst teams in better leagues are demoted while the best teams in weaker leagues are promoted. This article examines the economics of promotion and relegation, using data from English football (soccer). The crucial findings are as follows: players earn higher wages under promotion and relegation, promotion and relegation has a net positive effect on attendance, and the effect of promotion and relegation on competitive balance is ambiguous. The unbalancing effect arises because the system places some teams in leagues in which they have no realistic chance to afford a winning team, thereby causing teams to spend less on players during their (brief) stay in a higher league than they spent while trying to be promoted from as lesser league. The article concludes with an analysis of how promotion and relegation might be implemented in North America.

Unlike in the United States, most of the world's major professional sports leagues are not composed of a permanent roster of teams. Instead, major leagues promote and demote teams at the end of the season. The primary criterion for promotion and relegation is on-field success: The best teams in the highest ranking minor league are promoted to the major league, and the worst teams in the latter are reassigned to the former.[1] In larger countries, several lower leagues are organized hierarchically and the same promotion and relegation system applies down the line. In English football (soccer), for example, the hierarchy involves seven levels of leagues, and the bottom two levels are further divided into hierarchical divisions.

The primary difference between a system of leagues with fixed membership and a promotion/relegation system is that the latter permits a form of entry that is not feasible under the former. In leagues of fixed membership, entry occurs through expansion of a league or the formation of a new, competitive league. In the United

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States, expansion has been the primary means of entry into major leagues, although at some time during the 20th century the four most popular team sports experienced successful entry in that at least some of teams in a new league merged into the established league.[2] Entry and exit of leagues are substantially more common for minor league sports.

To prospective entrants, expansion and league entry have disadvantages. Expansion requires super-majority approval from established teams and entails paying a substantial entrance fee. Entry by an entire league requires that the incumbent league has left several viable franchise locations unexploited and that several entrants—not just one—are willing to suffer significant financial losses in the early years while the new league becomes established.

Promotion and relegation do not rule out these forms of entry, but this system makes other forms of entry feasible. First, an entrepreneur can buy a minor league team, hire high-quality players and coaches, and earn promotion to a higher league. Second, an entrepreneur can form a new team, enter into the bottom league, and then gain promotion to higher levels. In both cases, after a few years of successful play a team can be promoted all the way to the major league, although in the case of de novo entry in England this strategy requires at least eight promotions and therefore takes at least a decade.

These forms of entry require no approval by, and no payment of expansion franchises to, existing major league teams.[3] The main disadvantages of this form of entry are that teams must start at the bottom rather than immediately enter at the league level that they seek and then must dominate lower leagues to gain promotion, which implies that they may have to field teams that are too good (and too expensive) for the leagues in which they play.

The purpose of this article is to explore how promotion and relegation affect the economics of a sports team and league. Two questions are theoretical: Under what conditions will leagues with promotion and relegation experience more entry? and How does promotion and relegation alter the business strategy of established teams and leagues? The other questions are empirical: Does promotion and relegation produce measurable effects on entry and operations? To answer the second set of questions, this article examines the history of English football.

To my knowledge, these issues have been addressed only in simultaneous research by Stefan Szymanski and his coauthors and the references therein (see Hall, Szymanski, & Zimbalist, 2002 [this issue] and the references therein). Ross and Szymanski (2000) present a theoretical model that predicts that promotion and relegation increase the incentive of teams to invest in team quality to the benefit of fans, and argue that U.S. leagues should be required to adopt promotion and relegation through antitrust action against these leagues for their practices regarding team territorial rights, expansion, relocation, and stadium financing. The economic arguments of Hall et al. (2002) have many similarities with those advanced in this article; however, this article takes a somewhat different approach and addresses some different questions.

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THEORETICAL ISSUES

Entry in team sports has two distinct components. One is entry of a team into a specific location, and the other is a net increase in the number of teams that play in a league. Although the second type of entry requires the first, the first does not require the second. That is, a team can enter a particular location by changing the place it plays its home matches (relocation), or a new team can enter one location while an old team in another location exits the industry. For example, baseball's announced contraction plans may be combined with plans to enter lucrative new areas through expansion rather than relocation of its weakest franchises. Note that both relocation and promotion/relegation require entry and exit in different local markets. Alternatively, entry can cause a net increase in the number of teams (expansion, new leagues).

Promotion and relegation are a substitute for either relocation or expansion. In the first case, a higher league enters a new city (the home of the promoted team) while simultaneously exiting the home of the demoted team, whereas the lower league does the opposite. In the second case, a higher league promotes more teams than it demotes. If every league in the hierarchy follows this practice, the effect is net entry for the sport through the creation of new teams at the bottom of the hierarchy of leagues. Likewise, leagues can demote more teams than they promote as a means of contraction.

Team Strategies

The strategic decisions of a team are first whether to enter or exit the industry and then, conditional on entering (or not exiting), the level in the hierarchy of leagues to seek. The tactical decisions include picking the optimal team quality—that is, acquiring players and coaches of appropriate skill—to achieve the target league. Although the sequence of decisions is first to make the entry/exit decision, then the decision about league level, and finally a sequence of annual team quality decisions, logically each earlier decision is based on expectations about the financial consequences of later decisions, so it makes sense to discuss decisions about league attainment and team quality before examining the entry decision.

Team quality. The effects of promotion and relegation on team quality depend on how movement between leagues affects team profits. In principle, promotion and relegation can affect profits in two ways: A team's revenues and costs may differ in different leagues, and the presence of relegation and promotion may affect the interest of fans (and hence revenues).

Conceivably, good and bad teams could benefit from the promotion and relegation system. For the best teams, the reason is clear: Promotion is an additional reward for winning, and so conceivably can increase fan interest in games, assuming that fans prefer that their home team compete at a higher level. For the worst

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teams, the prospect of relegation also conceivably could increase interest because more is at stake in late-season games and because fans expect that in the next season the average quality of opponents may be lower. Whether these effects are present, of course, is an empirical matter, but if they are then sports that make use of promotion and relegation will earn higher revenues.

The other effect of promotion and relegation on team profits has to do with whether a team actually is better or worse off financially by being promoted to a higher league. To understand why promotion may not be profitable requires delving into the standard economic theory of a sports league. A sports team operates in two types of markets: a local market for selling tickets and concessions and a national or even international market for selling broadcast and licensing rights. In both cases, the demand for a team's products depends on its absolute quality, the quality of its opponents, and the team's tradition (or playing history). In addition, especially for local products, demand depends on exogenous demographic attributes of the city in which the team plays home games. In general, teams in more populous, wealthier markets will generate more revenue, holding constant the quality of the team and its opponents and the number and quality of its local competitors. All else being equal, a team in a better market will have a higher marginal revenue product of increments to team quality, so that teams that play in the best locations generally have a higher optimal quality than teams in worse locations.

Ignoring the possibility that the existence of the promotion and relegation system increases demand and taking as exogenous the number of teams in each league, the optimal distribution of teams among a hierarchy of leagues is for the best league to contain the teams that face the most intensive demand for the sport and for each league in the hierarchy to contain the teams with the best markets among those remaining after higher leagues have been filled. A promotion and relegation system forces a departure from the optimal allocation of teams at least half the time (and most likely virtually all of the time), given the unavoidable uncertainty associated with won-lost records during a league season as well as the requirement to relegate some teams even if the composition of the league is optimal. Thus, promotion/relegation inevitably has costs in terms of sportwide profitability.

Promotion and relegation reduce total revenues in a sport in the following way. When a team in a better market is demoted while a team in a worse market is promoted, all teams except the promoted team are likely to suffer financially. The demoted team will have an optimal relative quality in the lower league that is stronger than the team that was promoted, and so its demotion will reduce the expected won-loss record and hence the revenues of every team in the lower league. Likewise, the promoted team will have a lower optimal relative quality in the higher league than the team that was demoted. Hence, every other team in the higher league must play one home game with a team that has lower average quality than the team it replaced, causing each to have lower revenues. In both leagues, therefore, promotion/relegation reduces outcome uncertainty (competitive balance) and thereby reduces the demand for the other teams.

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Although promotion/relegation inevitably creates a cost, this cost could be small. That is, if the differences in local market demand are small among a large number of teams, the revenue loss from distributing teams suboptimally is likely to be small and may be offset by the demand-enhancing effects of promotion and relegation as described above. Indeed, the most plausible explanation for the widespread adoption of promotion/relegation systems, given that they are certain to cause a suboptimal distribution of teams among leagues, is that the demand-enhancing effect is present and that differences in market conditions are small among lower echelon teams in a stronger league and higher echelon teams in a weaker league.

Although theory does not yield a robust qualitative prediction about the effect of promotion on a team's profits, a team in a lower league may have an incentive to seek promotion to a higher league than the one to which it would be assigned if the sport maximized joint profits. Whether such an incentive exists depends on two factors. One factor is the possibility that a promotion and relegation system causes an increase in the demand for games involving teams that are candidates to change leagues, as discussed above. The other factor is the effect of promotion on team-specific revenues and costs.

In general, promotion into a league that is too strong for the market in which the team is located will have two economic effects. On the plus side, because the team's opponents are better and the higher league receives more attention from the press (basically, free advertising), the demand for the team's products should increase. On the negative side, the optimal quality of the team will be higher in absolute terms but lower in relation to other teams in its league. Promotion increases the marginal revenue product of quality and so causes the team to spend more on players if it joins a higher league. But a team from a market that, under an optimal assignment of teams to leagues, would be assigned to the lower league will have a higher optimal relative quality in the lower league than its optimal relative quality in the better league. Thus, a team that should be in a lower league from the perspective of sportwide profit maximization expects to finish higher in the standings in the lower league but spend more money for players in the better league.

Whether promotion is profitable to a team, then, depends on the net effect of all of these factors. Specifically, are the benefits of promotion (more revenues from playing better opponents, more exposure, and the possible demand-enhancing effects of the prospect of promotion in the year in which promotion is earned and of demotion later when the team returns to its optimal league) greater or less than the costs (higher salaries and a lower finish in the better league)? As a practical matter, this question is empirical, not theoretical, but the answer almost certainly is in the affirmative, for teams never refuse to be promoted.

The last important effect of promotion/relegation is on the overall quality of play. This effect is apparent from a simplified model of a sports league, as initially developed by El Hodiri and Quirk (1974). Assume that teams face revenues, R, and costs that are a function of quality, q, that the revenue function, $R(q)$, displays

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diminishing returns to quality, that is, that $R'(q) > 0$ and $R''(q) < 0$, and that the cost function is simply the amount of quality times its unit cost, or wq.[4] Finally, assume that teams maximize net revenues.[5]

In a league of fixed membership, the team solves this problem separately each year, which can be written as

$$\underset{q}{Max}\{R(q) - wq\} \tag{1}$$

and which yields the familiar first-order condition that marginal revenue equals marginal cost:

$$R' - w = 0. \tag{2}$$

Note that this equation implies competitive imbalance among teams that operate in markets with different demand intensity.[6] Suppose that at average team quality, q^*, Team 1 has more intense demand for team quality than Team 2. In this case, net revenues will increase if a unit of player quality is transferred from Team 2 to Team 1. To obtain equilibrium in Equation 2, the better teams need to acquire more quality than teams in weaker markets to drive the marginal revenue product of quality to its unit cost in every market.

In a promotion/relegation system, this year's decision about team quality affects the league in which the team plays next year. Assume that the team has a probability p of being promoted and a probability r of being relegated, with $p' > 0$ and $r' < 0$, and that q^*, q^p, and q^r represent, respectively, expected optimal team quality if the team stays in the same league, is promoted, or is relegated. Then, letting i represent the rate of time preference, the team's profit maximization problem for this period must take into account these contingencies and becomes

$$\begin{aligned} &Max\ R(q) - wq + (1-i) \\ &\left\{p\left[R^p\left(q^p\right) - wq^p\right] + r\left[R^r\left(q^r\right) - wq^r\right] + (1-p-r)\left[R(q^*) - wq^*\right]\right\}. \end{aligned} \tag{3}$$

The first-order condition for this problem is

$$(R' - w) + (1 - i)(p'[R^p(q^p) - wq^p] + r'[R^r(q^r) - wq^r] - (p' + r')[R(q^*) - wq^*] = 0. \tag{4}$$

This equation contains the difference between marginal revenue and marginal cost this season (as above) plus the financial consequences of promotion and relegation. If playing in higher leagues is more profitable then the second term is positive,

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which means that the quality level that solves Equations 1 and 2 is too small to solve Equations 3 and 4. Thus, holding constant market conditions and team locations, teams seek higher quality under promotion and relegation than they would seek in leagues of fixed size if teams are more profitable in higher leagues.[7] In the top league, teams at the top of the standings face a very low value of r and $p = 0$, so the prospect of promotion and relegation has virtually no weight in Equation 4. Hence, the equilibrium spread in team quality in the top league is lower than it would be in a league of the same teams with fixed membership. In other words, the effect of promotion and relegation is to increase competitive balance in the top league among the teams that are members; however, because the teams in the top league may not be the optimal allocation of teams to that league, this effect could be offset by the presence of teams in weak markets that have low optimal relative quality in the top league.

The extent to which promotion/relegation improves the overall quality of play depends on the supply of inputs for all teams in the market. The assumption that the supply of inputs is not perfectly inelastic is unexceptional for all inputs except players. For players, supply elasticity has two elements. One component is related to whether children decide to devote themselves to developing their athletic skills and then, as teens, to launch a career in sports. Presumably this elasticity is tiny, especially in the short run. The other element is the nationwide supply of existing players to professional teams. Because soccer is the primary sport in much of the world (most of Europe, Latin America, and Africa), even for a soccer power like England the supply of professional soccer talent is likely to be reasonably elastic, although not perfectly so if players prefer to play in their home country. Hence, the preceding simple model yields two important conclusions:

1. Holding the demand for soccer constant across countries and team locations within those countries, a nation with promotion/relegation will have stronger teams than a nation with leagues of fixed size. For example, the primary soccer league in the United States, Major League Soccer (MLS), always will be weaker than the top European leagues even if soccer becomes as popular in the United States as it is in Europe because MLS does not practice promotion and relegation.
2. The adoption of a promotion/relegation system will increase the wages of players as long as the world supply of professional soccer players is not perfectly elastic. This effect is not due solely to the possibility that promotion/relegation may increase demand but is a consequence of the fact that expenditures on players today are an investment in that they are a cost of gaining entry into a better league (or of avoiding demotion to a lower league), which causes this year's players to have greater value than they would without promotion and relegation.

Entry and exit. The preceding model has implications with respect to the simultaneous entry of one city and exit of another in a league. Recall that promotion/relegation generates more of this form of entry and exit than does voluntary relocation in a league of fixed size. A team moves from one location to another while staying in the same league because it believes that the new location is sufficiently more

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profitable that it will offset the costs of relocating. Because the cost of quality is the same in all locations, the source of profit differentials is differences in the intensity of demand. Hence, team relocation in closed leagues tends to be from weaker to stronger markets. By contrast, in a promotion/relegation system, a new city is added to a higher league because the team in that city does well on the field and perceives its profits to be enhanced by joining a better league, whereas the new city in the lower league is added coercively—the entrant has been forced to exit a more profitable league because of its poor on-field performance. In at least some cases, the demoted team would not want to relocate to the city of the team that replaced it through promotion/relegation.

Another form of entry into a league, whether from de novo creation of a new team or a permanent change in the league assignment of an existing team, arises when the entering team operates in a more lucrative local market than the worst existing team. Exogenous changes in the geographic distribution of people and wealth affect team revenue functions and so alter the set of optimal locations for teams in a league. In a world of perfect information, incumbent teams in a league would be able to identify these locations, and teams could relocate as soon as demographic shifts changed the list of optimal team locations. But if local market responsiveness in areas without a team is uncertain, local entrepreneurs plausibly can detect a favorable shift in local demand before teams that are located elsewhere. Under these conditions, an old team is likely to start to improve or a new team is likely to be formed in a growing local market before another existing team detects that the area is more profitable than its current location.

The preceding argument provides an unobvious economic explanation for a phenomenon that, among sports aficionados, is regarded as a cultural difference between North America and the rest of the world: In other nations, major league teams virtually never relocate. The absence of team movements is attributed to the deep roots that a team has in its local community and, as a result, the unacceptability of relocation to its fans—even in the United States, fans have deeply resented some relocations (e.g., the Brooklyn Dodgers and Cleveland Browns). In any case, this observation, if true, can be interpreted as an assertion that in most of the world relocation costs are higher relative to the costs of forming a new team than they are in the United States. De novo entry can be cheaper under promotion/relegation for two reasons: Expansion fees are avoided and information costs associated with assessing new markets may be lower for residents of an emerging market. Hence, the absence of team relocation is a plausible consequence of a system of promotion and relegation.

League strategies. The principal strategic decision of a league is to determine its size. The decision to expand is a response to growing demand, which makes an increasing number of locations viable franchise sites. But expansion is not usually in the interests of the most successful teams. Expansion entails increasing the number of teams in smaller markets that have no serious chance of winning a champion-

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ship. Hence, on average, expansion reduces the average revenue of teams because it reduces the average market potential of teams and hence the average quality and consequentiality of matches. Moreover, expansion requires adding competitors in national markets like broadcasting and licensing or, if national rights are pooled by the league, sharing the revenues from these rights among more partners. Thus, if expansion teams have, on average, weaker demand than incumbents, expansion reduces the national revenue of incumbents.

For the most part, leagues expand not because doing so is in the best interests of incumbents but because they are forced to do so. One purpose of expansion is to inhibit entry by competitive leagues. Another purpose is to satisfy demands from political officials in return for their support for favorable policies regarding sports, such as favorable tax treatments, antitrust exemptions, and stadium subsidies.[8]

New leagues are formed to take advantage of two sources of excess demand for professional sports. First, monopoly leagues usually do not place multiple teams in areas in which demand is sufficient to support more than one or two teams and, by setting monopoly prices, undersupply demand for products in national markets, such as television broadcasts. Second, monopoly leagues typically do not place teams in all financially viable franchise locations.

In leagues of fixed size, both types of excess demand arise for three reasons. First, teams either have exclusive territorial rights or, collectively, teams in areas that could support entrants can veto territory-invading expansion or relocation if either requires a supermajority vote of incumbent teams. Second, incumbent leagues set a monopoly price for an expansion team that exceeds that which is justified by the revenue potential of some otherwise viable locations. Third, relocation costs and imperfect information prevent efficient team relocation and league identification of all financially attractive expansion sites. In deciding on expansion, then, leagues seek to balance the benefits to incumbents of expansion and a lower probability of successful entry by competitive leagues against the cost of adding to the number of teams in weak markets or increasing competition in strong markets. Inevitably, this calculation leads a monopoly league to be smaller than the number of teams that would exist under competition or the absence of league-created entry barriers.

The promotion/relegation system reduces pressures to expand by providing a mechanism whereby teams in markets that have become stronger due to demographic shifts can enter an established league. Promotion/relegation has two benefits to incumbents: The system creates a barrier to entry by new leagues into unoccupied attractive markets (a team in a lower league is already there and likely to seek promotion itself if the market warrants it) and it provides a mechanism for limiting the total number of teams in the league that play in weak markets. Whereas this system denies incumbents the opportunity to charge expansion fees, it compensates them by not forcing them to play more games against weak teams and to share national markets with more teams. But regardless of whether promotion/relegation

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is more or less profitable for the best teams than expansion, the one unambiguous prediction is that leagues using this system ought to be less inclined to expand.

The last effect of a decision to adopt promotion/relegation pertains to its effects on player costs. Abstracting from the demand-enhancing effects of the system, leagues have a good reason to adopt fixed membership—doing so reduces equilibrium quality and, if player supply is not perfectly elastic, player wages. Yet all major European soccer powers use promotion/relegation, which seems to be at odds with the assumption that leagues maximize profits.

Two explanations for this phenomenon are consistent with the standard economic model of a for-profit sports league.

First, until very recently, the world market for soccer players was monopsonized. Players were signed to their first contracts while still in their midteens, sometimes by an amateur or semiprofessional team that was owned and operated by the professional team in their home neighborhood. Subsequently, if a player switched teams after a contract expired, the new team was forced to pay compensation—a transfer fee—to the old team. For strong professional players, transfer fees typically ran in the millions of dollars, which was an order of magnitude more than the players' salaries. These fees applied to international transfers as well as movements within the same country and were enforced by the international governing body of soccer, the Federation Internationale de Football Associations (FIFA). Thus, secure monopsonization of the player market ensured that at least some of the demand-enhancing effect of promotion/relegation would be passed on to teams.

This explanation for promotion/relegation may have disappeared. In December 1995, the European Court held that the transfer fee system violated the single-market principle by inhibiting free movement of labor (see Jeanrenaud & Késenne, 1999). As a result, European players can move freely within Europe after their contracts expire. The industry is seeking special legislation from the European Union to allow the monopsony in the player market to be reinstated, but if this fails then the transfer fee rules throughout the world are very likely to disappear, causing the market for the best players to become competitive.

A second explanation for the dominance of promotion/relegation arises from the fact that an extremely lucrative part of professional soccer is international matches, especially between the best teams in the top leagues. In Europe, the top teams in national Division 1 leagues qualify for the Champions League in the following season, which involves playing several very lucrative matches. Leagues want their best teams to be competitive in the Champions League to make participation in these matches more profitable. Hence, it makes sense to adopt a system that, all else equal, produces stronger teams. Thus, adoption of promotion/relegation is much like an arms race: Conceivably, all teams and top leagues would be better off without it, but once one nation adopts the system, the others have a financial incentive to follow.

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THE ENGLISH NATIONWIDE FOOTBALL LEAGUE

England is widely regarded as having one of the most successful sports leagues in the world—the English Premier League. For the league as a whole, average revenues per team in 1999-2000 exceeded $50 million, and average accounting profits per team were about $5 million (for data sources, see the appendix). At present, 10 members of the Premier League and 7 members of Division 1 are publicly traded corporations. Together, these 17 teams had a market capitalization in 2000 of more than $3 billion, or about $200 million each. The market cap of the most successful team, Manchester United, hovers around $1 billion, making it among the two or three most valuable sports franchises in the world. Thus, the English Premier League is roughly comparable to Major League Baseball (MLB), the National Basketball Association (NBA), and probably behind only the National Football League (NFL) in profits per team and average franchise value, while its best teams are comparable to the best in any league, including the NFL.

The basic institutional structure of English football has been stable for several decades. As in all countries, the overall governing body for English soccer is FIFA, located in Switzerland. FIFA determines the eligibility of teams and players for international competitions, such as the World Cup, and insists that internationally sanctioned teams must play only other sanctioned teams or else they and their players lose eligibility for international competition. Because international competition is very important to players and teams, this sanctioning power enables FIFA to regulate the business activities of soccer.

Below FIFA is the Union of European Football Associations (UEFA), which sets rules for European Football that are not dictated by FIFA, organizes international competitions within Europe, and supervises national football organizations. The Football Association (FA) represents all of the British Isles in UEFA, including Ireland as well as England, Northern Ireland, Scotland, and Wales. Each of these political jurisdictions has its own top (Division 1) league, its own hierarchy of minor leagues, and its own national team for competing in the World Cup, the Olympics, and other international championships. The FA sanctions and governs all professional and amateur football leagues and teams in the British Isles.

The Premier League stands at the top of the hierarchy of English leagues. Below it three other leagues collectively are called the Nationwide Football League and individually were called Divisions 2, 3, and 4 until 1992 but since have been called Divisions 1, 2, and 3. Historically, all four leagues collectively were called the Nationwide Football League until 1983, when the leagues began to sell their names to corporate sponsors. The leagues were known as the Canon League Divisions 1 through 4 from 1983-1986, the Today League in 1986-1987, and the Barclays League from 1987-1992, when the Premier League decided to withdraw from the Nationwide Football League. In the 1992-1993 season, the lower leagues became Divisions 1, 2, and 3 and retained the name Barclays League, whereas the top

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league became the English Football Association Premier League. In 1993, the top league sold its name to Carling and has been known as the Carling Premier League since. Meanwhile, Divisions 1 through 3 were the Endsleigh League from 1993 to 1996 but have been just the Nationwide Football League since then. To avoid confusion, the names of leagues herein will be simply Premier and Division 1, 2, and 3 for all historical periods.

Beneath these leagues are a large number of lesser clubs and leagues. The highest ranking of the remaining leagues is the Nationwide Conference. Several teams in the Conference have played in the Nationwide Football League and even the Premier League in the past. Although teams in the Conference and below must not be fully professional they can pay players, and many aspire to membership in the top four leagues and so employ several players who are sufficiently talented that they could play for a team in one of the top four professional leagues. Below the Nationwide Conference are three leagues of roughly equal stature: the Unibond League, the Dr. Martens League, and the Ryman Football League. Unibond has two divisions (premier and first), Dr. Martens has three (premier and coequal eastern and western divisions), and Rymans has four (premier, first, second, and third divisions). Finally, below these leagues are 22 "nonleague" leagues, several of which are further divided into hierarchical divisions. As one proceeds down the list, rules about paying players become increasingly restrictive.

At each stage of the hierarchy, leagues engage in promotion and relegation. In addition to promotion and relegation between adjacent pairs of the four top leagues, since 1987 Division 3 usually has exchanged one team per year with the Nationwide Conference while the latter drops three teams each year and accepts one team each from the premier divisions of Unibond, Dr. Martens, and Ryman. These leagues, in turn, exchange teams with the leagues below them, and the lowest ranking divisions in these three leagues exchange teams with the top regional "nonleague" leagues.

Entry

The purpose of this section is to examine entry and exit in the top four English leagues to see if this history supports the theoretical arguments about the effect of promotion and relegation on entry into a major league and to determine the extent to which promotion and relegation cause instability in league membership through entry and exit. To summarize, the core predictions of the theory of a sports league are that the system of promotion and relegation reduces the rate of expansion and the entry of new major leagues but provides opportunities for new teams to enter or for teams in more rapidly growing markets to improve their quality and earn promotion while teams in declining markets gradually descend the hierarchy of leagues.

Unfortunately, the effect of promotion and relegation on the entry of new leagues is not observable because of the institutional rules of football. In England, as elsewhere, entry of top-ranking leagues is essentially impossible in the present

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environment. One of FIFA's rules is that if a national team or any professional and amateur team wants to compete internationally, whether in major championships or just friendly (exhibition) matches, it must have a national governing body that enforces FIFA rules, grants franchises to leagues and teams, and certifies the eligibility of players. Another FIFA policy is that each country should have only one major league, usually called Division 1, although in England it has been called the Premier League since 1992. In the United States, MLS is the designated Division 1 league. Thus, entry of competing major leagues is precluded because FIFA and the English soccer authority are unlikely to sanction one. Nevertheless, FIFA and its subsidiary bodies (including the FA and the U.S. Soccer Federation) do allow entry of competing minor leagues. In England, monopoly leagues occupy the first five levels of the hierarchy, but the lower levels have competitive leagues.

An important aspect of the history of English football is that neither leagues nor the FA recognizes territorial rights. Teams can form anywhere, can move their home venues, and can be promoted into a league with a nearby local competitor. The only limitations are league rules regarding the suitability of a playing site in terms of the size and quality of the field and stadium. Because territorial rights are not enforced, large cities have many teams. From 1990-1991 to 2001-2002, 18 of the 36 teams that have played in the Premier League come from multiteam markets. London alone is home to 10 teams that played in the Premier League during this period, and Liverpool, Manchester, Nottingham, and Sheffield have had 2 Premier League members each. By contrast, the much larger United States has only four multiteam markets in the major leagues (and none in MLS): New York in all four major sports (baseball, basketball, football, and hockey), Los Angeles in three sports, San Francisco in two sports, and Chicago in one sport. Only New York has three teams in one sport (hockey). The absence of territorial rights reduces the "big market, small market" demand disparity; however, this check is limited because the most successful teams enjoy some immunity to competition due to the fan attachments that they have accumulated during their histories.

Early in the history of English football the primary mechanism for entry was the creation of new leagues to increase the number of levels in the hierarchy, but expansion of existing leagues also was important. The Premier League was organized in 1888 as a league of 12 teams, and in 1891 it expanded to 14. In 1892, Division 1 entered with 16 teams, and promotion and relegation began at the end of its first season. In 1893 and 1894, 2 Division 1 teams were promoted to the Premier League while only 1 team was relegated, thereby expanding the Premier League to 15 and then 16 teams. In 1893 and 1894, Division 1 added 1 team to retain its size of 16. In 1898, both leagues expanded to 18. The Premier League did so by promoting 2 teams while relegating none, whereas Division 1 simply expanded by 4. In 1905, both leagues expanded to 20 in the same fashion. World War I interrupted play from 1915 to 1919, but when play resumed both leagues expanded to 22 teams, a gain of 4 per league. This expansion was the last for the upper echelon of English football, for the combined size of these two leagues has remained at 44 teams ever since.

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Division 2 was created in 1920 with 22 teams, and then in 1921 expanded to two leagues, a northern branch of 20 teams and a southern branch of 22. Both leagues were included in the promotion/relegation system as the league champion in each branch was promoted while the 2 worst teams in Division 1 were demoted. Two years later, the northern branch expanded to 22 teams. Thus, in 1923, the top four English football leagues included 88 teams, compared to 92 today. In essence, the size of top professional football in England has remained roughly the same for more than three fourths of a century.

Except for 1931-1932, when the north branch of Division 2 had 21 teams after one of its members folded, all leagues remained at these sizes until after World War II. Play again was suspended during the war, from three games into the season in 1939 to the fall of 1946. When play resumed, all four leagues reentered with exactly the same size and composition. The last expansion was in 1950, when the two branches of Division 2 expanded to 24 teams each and the Nationwide Football League reached its current size of 92 members.

Of the 92 positions in the top four English leagues that were created between 1888 and 1950, 70 slots arose from entry of new leagues and 22 represent expansion of existing leagues. For more than 50 years thereafter, no net entry has occurred, either from new leagues at this level of quality or expansion of existing leagues. The economic growth of the past 50 years, the arrival of television with its new major source of revenues, the creation of lucrative international competitions for professional teams, and the boom in sports since the mid-1970s have substantially increased the demand for English football and the revenues of the teams. But this growth in demand has not led to growth in the number of teams in the top four professional leagues. By contrast, the number of teams in the four major professional sports in the United States has more than doubled in this period. Nine new leagues have entered, six of which have survived by being merged into the established league, and all four leagues have added expansion teams every few years since the 1960s. Thus, for more than half a century, the behavior of the top four leagues in England and the four American major leagues is consistent with the theoretical prediction that a system of promotion and relegation reduces the propensity of new leagues to enter and old leagues to expand.

Two major changes since 1950 have diminished effective quality-adjusted supply. The first was the restructuring of Division 2 in 1958, when Division 3 was created by demoting half the teams in the two branches of Division 2. The second was the contraction of the Premier League from 22 to 20 teams. The Premier League shrunk from 22 teams in 1986-1987 to 21 in 1987-1988 to 20 in 1988-1989, but then in 1991-1992 it temporarily returned to 22. Then, after it withdrew from the Nationwide Football League, the Premier League shrunk back to 20 in 1995-1996, where it remains today. These changes were undertaken by promoting and demoting different numbers of teams.

The main benefit of promotion and relegation is that it enables entry of teams into high-quality leagues; however, extensive entry is not guaranteed. Instead, the

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same small core group of teams could simply rotate membership in the leagues, with neither entry nor exit among this group. The following review of the history of promotion and relegation practices in England shows that a core group of teams tends to dominate the Premier League but that the system of promotion and relegation has produced considerable instability in about one third of Premier League membership and in the membership of lower leagues.

The effect of the promotion/relegation system has varied greatly during the history of English football because the number of teams promoted each year has changed many times. The standard practice from 1893 through 1970 was to demote 2 Premier League teams and to promote 2 Division 1 teams each year. Since the 1970s, the standard practice has been to promote and demote 3 teams. Likewise, the number of teams demoted from Division 1 and promoted from Division 2 initially was 1, but when two Division 2 leagues were created, Division 1 promoted the winner of both Division 2 leagues. After Division 2 placed half its teams in Division 3, promotion and relegation between Divisions 1 and 2 involved 3 teams from each league. When Division 2 north and south became Divisions 2 and 3, these leagues immediately began to promote and relegate 4 teams each year.

Another important changing feature of English football is the mechanism for relegating teams out of Division 3. From the formation of the Premier League in 1888 through the 2001-2002 season, 127 different teams have played in at least one of the four top leagues. Of these, two losses from the roster of teams were due to mergers, so 94 of the 127-member teams are now in one of the four leagues. Thus, the number of teams that have entered and exited the four-league structure is 33 during 103 seasons, or fewer than 1 new team every 3 years.

In its first 45 years (through 1932), the number of teams that left the four-league structure was 18, or 1 team every 2.5 years. Then, between 1932 and 1950, no teams left the four leagues. Thus, the four Nationwide Football League divisions were a closed league between 1932 and 1950, which is also the last year in which the Nationwide League expanded.

Beginning with the 1950-1951 season, formal promotion and relegation occasionally took place but not on a regular basis. Division 3 does not automatically relegate and promote a team every year. Instead, the members of the Nationwide Football League decide whether to demote a team and promote another at the end of the season, based on on-field and financial performance and the suitability of a team's venue. In some cases the demoted team is not the team that finished at the bottom of Division 3.

The first postwar relegation from Division 3 occurred in 1951, when New Brighton was relegated. The next demotion took place in 1960, with the loss of Gateshead. During the 1961-1962 season, Accrington Stanley withdrew from the league after 33 games (of a scheduled 46). Thus, between 1950 and 1970, 3 teams left the Nationwide League, or 1 every 7 years, which is a much slower rate of turnover than occurred between 1888 and 1932.

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Relegation from Division 3 became more common in the 1970s. Bradford Park Avenue, which had once been a member of the Premier League, was removed from Division 3 in 1970, and additional teams were removed in 1972 and 1977. After another hiatus, 1 team per year was removed from 1987 until 1990, then none in 1991. In 1992, 2 teams voluntarily withdrew from Division 3, followed by 1 relegation in 1993. After the Premier League declared its independence from the other leagues and decided to contract, relegation from Division 3 ceased during the ensuing adjustment period, but annual relegation began again in 1997. Thus, between 1951 and 2001, 18 teams were promoted to Division 3 from the Nationwide Football Conference, of which 15 took slots made available by relegation and 3 replaced a squad that withdrew. Twelve promotions have occurred since 1987, or about 4 every 5 years, which is the fastest turnover rate by simultaneous entry and exit since the 1920s and early 1930s.

The effect of entry in the postwar era on the upper two leagues has not been very substantial. Including the 4 expansion teams of 1950, 22 teams have been promoted to Division 3 since 1950 (including Colchester United twice, for a total of 23 promotions). Among these, only Wimbledon, which was promoted to Division 3 in 1977, has ascended to the Premier League. Five of the 21 have been promoted to Division 1. Among the 1950 expansion teams, 3 (Gillingham, Scunthorpe, and Shrewesbury) played several years in Division 1, whereas Colchester United has divided its time roughly equally between Divisions 2 and 3 (with 2 years, 1990-1992, in the Conference). Peterborough United (1960) and Oxford United (1962) also have ascended to Division 1. No team that has been promoted from the Conference after 1977 has played in Division 1.

Nine of the 21 teams that have entered since 1950 have been relegated back to the Conference. Barnet, Hereford United, Maidstone United, Scarborough, and Workington were relegated from Division 3 and never returned, whereas Colchester United, Darlington, Halifax Town, and Lincoln City were relegated and then promoted back. These 9 and 7 other teams that have been promoted to Division 3 have either stayed there or peaked at Division 2, although 5 of these have entered since 1997 and so have not had much time to ascend the hierarchy.

In half a century, promotion and relegation between Division 3 and the Nationwide Conference has brought 19 teams into the Nationwide Football League. The net effect of the promotion/relegation system has been to change the identities of 13 of the 92 members of the top four leagues because 3 teams quit Division 3 voluntarily and 3 more were relegated and then promoted back. Ten of these 13 additions thus far involve switches in the composition of only the bottom two leagues.

The teams that have been promoted to Division 3 from the Conference are not recent entrants into football. Of the 92 teams in the top four leagues in 2001-2002, 71 were formed in the 19th century. The youngest teams are Colchester United (formed in 1937), Peterborough United (formed in 1934), and Wigan Athletic (formed in 1932), although several have undergone reorganizations or mergers

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since they were founded. Wimbledon, which has the most recent rise from entry into Division 3 in 1977 to Premier League membership in 1992, was founded in 1889 as an amateur squad and then began its climb up the hierarchy when it became professional in 1964, rising to Division 3 in 13 years and the Premier League in 15 more.

The most recent entrant to Division 3, Rushden and Diamonds, reveals the more typical pattern of team improvement. Rushden Town was formed in 1889 and Irthlingborough Diamonds was formed in 1947. In 1992, the teams merged under a new leader, Max Griggs of the R. Griggs Group that manufactures Dr. Martens footware and sponsors a minor league that feeds the Nationwide Conference. The team then began its ascent up the hierarchy, reaching Division 3 in 9 years.[9]

The remarkable aspect of the nature and effects of entry into the Nationwide Football League is that it implies relatively little change in the optimal composition of the four top professional leagues for a half century and virtually no change in the optimal composition of the top two leagues. Examination of the histories of the older teams, however, reveals substantial change in the composition of the four leagues among the 70 teams that have belonged to the Nationwide Football League continuously since the end of the war.

The history of the Premier League reveals two core facts. First, a few teams have rarely spent any significant time out of the Premier League and account for most of its championships. Second, there is considerable turnover in Premier League membership among teams that are not frequent challengers for the Premier League championship.

Premier League championships have been relatively concentrated. A total of 61 teams played in the Premier League between 1888-1889 and 2000-2001, and 23 have won the championship. Whereas only about one third of the teams that played in the Premier League and one sixth of all Nationwide Football League members have won a championship, the number of different champions exceeds the current membership of the league (20) and its peak membership (22).

Because of war interruptions, the Premier League crowned 102 champions in the 113 years between 1888-1889 and 2000-2001. Eight teams—Liverpool (18), Manchester United (14), Arsenal (11), Everton (9), Aston Villa (7), Sunderland (6), Newcastle United (4), and Sheffield Wednesday (4)—account for more than 70% of these championships. Moreover, all of these teams have been in either the Premier League or Division 1 for nearly every season in the 20th century. Everton has spent 99 of its 103 seasons in the Premier League, having last played in Division 1 in 1953-1954, and Arsenal has played in the Premier League every year since 1919. Manchester United has missed 2 Premier League seasons since 1936.

The original 12 members of the Premier League have proven to be very durable participants at the top level of English football. Five were members of the Premier League in 2001-2002: Aston Villa, Blackburn Rovers, Bolton Wanderers, Derby County, and Everton. Four were in Division 1 in 2001-2002: Burnley, Preston North

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End, West Bromwich Albion, and Wolverhampton Wanderers. Two original members played in Divison 2 (Notts County and Stoke City), and 1 (Accrington) was disbanded in 1893 after 5 years in the Premier League.

Table 1 lists all members of the Premier League during its history. The 8 most frequent members were members of the Premier League or Division 1 in the 19th century. Of the 20 Premier League members in 2001-2002, 9 rank among the top 10 teams in total appearances in the league and 16 rank among the top 25. The 2001-2002 team with the fewest Premier League years is Fulham at 13, and the 2001-2002 member with the fewest number of years in the four top leagues is Ipswich Town with 57. Finally, of the 61 teams that have played at least once in the Premier League, only one no longer exists (Accrington). Three others are no longer in any of the top four leagues but play in lower leagues: Bradford Park Avenue (Unibond Divison 1), Darwen (First North Western Train League Division 2), and Glossop North End (First North Western Train League Division 1).

Table 2 shows the composition of the Premier League since the 1990-1991 season. A total of 36 teams have played in the Premier League during these 12 years, when the league varied in size between 20 and 22 teams. As is summarized in Table 3, the teams that played in the Premier League fall into three basic categories: a core of more or less permanent members, a group that is referred to as yo-yo teams because they move up and down with regularity, and a group of teams that only occasionally make the Premier League (and then for only 1 or 2 years).

In the first group are 9 teams that played in the league all 12 seasons since 1990 and 5 more that played between 9 and 11 seasons. Thus, of the 20 current slots in the Premier League, 13 slots (on average) are occupied by 14 teams that constitute the core membership. Another 12 teams spent between 4 and 8 years in the Premier League during this period, and these constitute the yo-yo group. Finally, 10 more teams spent 3 or fewer years in the Premier League. These teams tended to be very inconsistent in their play, as 8 of the 10 also played in Division 2 during the decade compared to only 1 of the yo-yo group. Indeed, 2 of these 8 (Fulham and Notts County) played in all four leagues in 12 years.

The top 10 teams in the 1990s have had a very long run. During the past 25 years, 5 of these teams (Arsenal, Everton, Liverpool, Manchester United, and Tottenham Hotspur) have played every season in the Premier League. Among the other 5, Aston Villa spent 1986-1987 in Division 1, Chelsea and Leeds spent several years in Division 1 in the 1980s, Southampton had a similar spell in the 1970s, and Coventry City was relegated to Division 1 in 2001 for the first time since 1966-1967.

The histories of the group of recent occasional Premier League teams reveals the most interesting aspect of entry and exit in the promotion/relegation system, which is the inconsistency of many of the teams.

The Bolton Wanderers were an original member of the Premier League in 1888 and were a fixture into the 1960s. With the exception of a brief visit to the Premier League in the late 1970s, the team then sank into the lower divisions, and as late as

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TABLE 1: English Premier League Composition, 1888-2002

Team	Number of Years		Team	Number of Years	
	Premier League	*Football League*		*Premier League*	*Football League*
Everton[a]	99	103	Blackpool	27	95
Aston Villa[a]	91	103	Ipswich Town[a]	26	57
Liverpool[a]	87	98	Portsmouth	26	75
Arsenal[a]	85	98	Bury	22	97
Manchester United[a]	77	99	Charlton Athletic[a]	21	74
Sunderland[a]	74	101	Norwich City	20	75
Manchester City	73	99	Queen's Park Rangers	20	75
Newcastle United[a]	72	98	Luton Town	16	78
Chelsea[a]	67	86	Cardiff City	15	75
Tottenham Hotspur[a]	67	83	Wimbledon	14	26
West Bromwich Albion	68	103	Fulham[a]	13	84
Sheffield Wednesday	66	99	Bradford City	12	88
Derby County[a]	64	103	Crystal Palace	12	75
Bolton Wanderers[a]	63	103	Grimsby Town	12	98
Blackburn Rovers[a]	62	103	Oldham Athletic	12	84
Sheffield United	59	99	Bristol City	9	90
Wolverhampton Wanderers	59	103	Watford	7	75
Nottingham Forest	56	99	Accrington[b]	5	5
Middlesbrough[a]	53	92	Brentford	5	75
Stoke City	52	85	Brighton/Hove Albion	4	75
Burnley	51	103	Bradford Park Avenue[c]	3	51
Birmingham City	50	99	Oxford United	3	40
Leeds United[a]	48	75	Darwen[c]	2	8
West Ham United[a]	47	76	Millwall	2	75
Preston North End	46	103	Swansea City	2	75
Leicester City[a]	45	97	Barnsley	1	93
Coventry City	34	76	Carlisle United	1	67
Southampton[a]	32	75	Glossop North End[c]	1	17
Huddersfield Town	30	81	Leyton Orient	1	86
Notts County	30	103	Northampton Town	1	75
			Swindon Town	1	75

a. Premier League members from 2001-2002.
b. Disbanded.
c. Now play in lower classification leagues.

1987-1988 was playing in Division 3. The team then revived, climbing back to regain Premier League membership in 1995-1996. Bolton is now a yo-yo team, spending 3 of the past 7 years in the Premier League and 4 in Division 1.

Notts County is another original member of the Premier League that spent most of its time in either the Premier League or Division 1 before World War II but after-

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TABLE 2: League Composition in British Football From 1990-1991 to 2001-2002

	Teams With Premier League Status											
Team	*1990*	*1991*	*1992*	*1993*	*1994*	*1995*	*1996*	*1997*	*1998*	*1999*	*2000*	*2001*
Arsenal	P	P	P	P	P	P	P	P	P	P	P	P
Aston Villa	P	P	P	P	P	P	P	P	P	P	P	P
Barnsley	1	1	1	1	1	1	1	P	1	1	1	1
Blackburn Rovers	1	1	P	P	P	P	P	P	P	1	1	P
Bolton Wanderers	2	2	2	1	1	P	1	P	1	1	1	P
Bradford City	2	2	2	2	2	2	1	1	1	P	P	1
Charlton Athletic	1	1	1	1	1	1	1	1	P	1	P	P
Chelsea	P	P	P	P	P	P	P	P	P	P	P	P
Coventry City	P	P	P	P	P	P	P	P	P	P	P	1
Crystal Palace	P	P	P	1	P	1	1	P	1	1	1	1
Derby County	P	1	1	1	1	1	P	P	P	P	P	P
Everton	P	P	P	P	P	P	P	P	P	P	P	P
Fulham	2	2	2	2	3	3	3	2	2	1	1	P
Ipswich Town	1	1	P	P	P	1	1	1	1	1	P	P
Leeds United	P	P	P	P	P	P	P	P	P	P	P	P
Leicester City	1	1	1	1	P	1	P	P	P	P	P	P
Liverpool	P	P	P	P	P	P	P	P	P	P	P	P
Luton Town	P	P	1	1	1	1	2	2	2	2	2	3
Manchester City	P	P	P	P	P	P	1	1	2	1	P	1
Manchester United	P	P	P	P	P	P	P	P	P	P	P	P
Middlesbrough	1	1	P	1	1	P	P	1	P	P	P	P
Newcastle United	1	1	1	P	P	P	P	P	P	P	P	P
Norwich City	P	P	P	P	P	1	1	1	1	1	1	1
Nottingham Forest	P	P	P	1	P	P	P	1	P	1	1	1
Notts County	1	P	1	1	1	2	2	3	2	2	2	2
Oldham Athletic	1	P	P	P	1	1	1	2	2	2	2	2
Queen's Park Rangers	P	P	P	P	P	P	1	1	1	1	1	2
Sheffield United	P	P	P	P	1	1	1	1	1	1	1	1
Sheffield Wednesday	1	P	P	P	P	P	P	P	P	P	1	1
Southampton	P	P	P	P	P	P	P	P	P	P	P	P
Sunderland	P	1	1	1	1	1	P	1	1	P	P	P
Swindon Town	1	1	1	P	1	2	1	1	1	1	2	2
Tottenham Hotspur	P	P	P	P	P	P	P	P	P	P	P	P
Watford	1	1	1	1	1	1	2	1	1	P	1	1
West Ham United	1	P	1	P	P	P	P	P	P	P	P	P
Wimbledon	P	P	P	P	P	P	P	P	P	P	1	1

NOTE: P = Premier League, 1 = Division 1, 2 = Division 2, and 3 = Division 3.

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TABLE 3: Concentration in Premier League Membership, 1991-2002

Number of Years in Premier League	*Number of Teams*	*Team Name*
12	9	Arsenal, Aston Villa, Chelsea, Everton, Leeds United, Liverpool, Manchester United, Southampton, and Tottenham Hotspur
11	1	Coventry City
10	2	Wimbledon and West Ham United
9	2	Newcastle United and Sheffield Wednesday
8	1	Blackburn Rovers
7	5	Derby County, Leicester City, Manchester City, Middlesbrough, and Nottingham Forest
6	1	Queen's Park Rangers
5	4	Crystal Palace, Ipswich Town, Norwich City, and Sunderland
4	1	Sheffield United
3	3	Bolton Wanderers, Charlton Athletic, and Oldham Athletic
2	2	Bradford City and Luton Town
1	5	Barnsley, Fulham, Notts County, Swindon Town, and Watford

ward sank all the way to Division 3. Since then it has ascended to the Premier League twice, once in the early 1980s and again in the early 1990s; however, between 1992 and 1998 it fell all the way back to Division 3, then climbed back to Division 2.

Watford spent most of the 1950s, 1960s, and 1970s in Division 2, then in the late 1970s began to improve until, in the mid-1980s, it played in the Premier League. Demoted to Division 1 in 1988, the team sank to Division 2 in 1997; however, after being purchased by pop star Elton John, the team climbed back for a 1-year stay in the Premier League in 1999-2000.

Table 4 summarizes the league memberships for the past 11 years of the teams that did not play in the Premier League during the 1990s. During this period, the three lower divisions contained between 72 and 74 teams. A total of 90 teams played in one of these divisions during the period, and 63 of these did not play in the Premier League. Only 4 of these 63 teams played in the same league for all 11 years: Portsmouth and Wolverhampton in Division 1, AFC Bournemouth in Division 2, and Rochdale in Division 3. Fifteen teams played in all three leagues, and 4 more played in Divisions 2 and 3 plus the lower minor leagues. The remaining 40 teams played in two leagues (possibly including the Nationwide Conference). By contrast, of the 36 teams that played in the Premier League during this period, 9 played every season in the Premier League and 17 others played only in the Premier League and Division 1. Eight teams played in three leagues and 2 played in four. Thus, the teams that make the Premier League tend to be more consistent in quality than the teams that either never or only rarely make the Premier League.

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TABLE 4: League Composition in British Football, 1991-2002

	Other Football League Teams										
Team	*1991*	*1992*	*1993*	*1994*	*1995*	*1996*	*1997*	*1998*	*1999*	*2000*	*2001*
AFC Bournemouth	2	2	2	2	2	2	2	2	2	2	2
Aldershot	3	a	a	a	a	a	a	a	a	a	a
Barnet	3	3	2	3	3	3	3	3	3	3	a
Birmingham City	2	1	1	2	1	1	1	1	1	1	1
Blackpool	3	2	2	2	2	2	2	2	2	3	2
Brentford	2	1	2	2	2	2	2	3	2	2	2
Brighton/Hove Albion	1	2	2	2	2	3	3	3	3	3	2
Bristol City	1	1	1	1	2	2	2	1	2	2	2
Bristol Rovers	1	1	2	2	2	2	2	2	2	2	3
Burnley	3	2	2	1	2	2	2	2	2	1	1
Bury	2	3	3	3	3	2	1	1	2	2	2
Cambridge United	1	1	2	2	3	3	3	3	2	2	2
Cardiff City	3	3	2	2	3	3	3	3	2	3	2
Carlisle United	3	3	3	3	2	3	2	3	3	3	3
Cheltenham Town	a	a	a	a	a	a	a	a	3	3	3
Chester City	2	2	3	2	3	3	3	3	3	a	a
Chesterfield	3	3	3	3	2	2	2	2	2	3	2
Colchester United	a	3	3	3	3	3	3	2	2	2	2
Crewe Alexandra	3	3	3	2	2	2	1	1	1	1	1
Darlington	2	3	3	3	3	3	3	3	3	3	3
Doncaster Rovers	3	3	3	3	3	3	3	a	a	a	a
Exeter City	2	2	2	3	3	3	3	3	3	3	3
Gillingham	3	3	3	3	3	2	2	2	2	1	1
Grimsby Town	1	1	1	1	1	1	2	1	1	1	1
Halifax Town	3	3	a	a	a	a	a	3	3	3	3
Hartlepool United	2	2	2	3	3	3	3	3	3	3	3
Hereford United	3	3	3	3	3	3	a	a	a	a	a
Huddersfield Town	2	2	2	2	1	1	1	1	1	1	2
Hull City	2	2	2	2	2	3	3	3	3	3	3
Kidderminster Harriers	a	a	a	a	a	a	a	a	a	3	3
Leyton Orient	2	2	2	2	3	3	3	3	3	3	3
Lincoln City	3	3	3	3	3	3	3	2	3	3	3
Macclesfield Town	a	a	a	a	a	a	3	2	3	3	3
Maidstone United	3	a	a	a	a	a	a	a	a	a	a
Mansfield Town	3	2	3	3	3	3	3	3	3	3	3
Millwall	1	1	1	1	1	2	2	2	2	2	1
Northhampton Town	3	3	3	3	3	3	2	2	3	2	2
Oxford United	1	1	1	2	2	1	1	1	2	2	3
Peterborough United	2	1	1	2	2	2	3	3	3	2	2
Plymouth Argyle	1	2	2	2	3	2	2	3	3	3	3
Portsmouth	1	1	1	1	1	1	1	1	1	1	1
Port Vale	1	2	2	1	1	1	1	1	1	2	2
Preston North End	2	2	3	3	3	2	2	2	2	1	1
Reading	2	2	2	1	1	1	1	2	2	2	2
Rochdale	3	3	3	3	3	3	3	3	3	3	3

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TABLE 4: Continued

	Other Football League Teams										
Team	*1991*	*1992*	*1993*	*1994*	*1995*	*1996*	*1997*	*1998*	*1999*	*2000*	*2001*
Rotherham United	3	2	2	2	2	2	3	3	3	2	1
Rushden & Diamonds	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	3 [a]
Scarborough	3	3	3	3	3	3	3	3			
Scunthorpe United	3	3	3	3	3	3	3	3	2	3	3
Shrewsbury Town	2	3	3	2	2	2	3	3	3	3	3
Southend United	1	1	1	1	1	1	2	3	3	3	3
Stockport County	2	2	2	2	2	2	1	1	1	1	1
Stoke City	2	2	1	1	1	1	1	2	2	2	2
Swansea City	2	2	2	2	2	3	3	3	3	2	3
Torquay United	2	3	3	3	3	3	3	3	3	3	3
Tranmere Rovers	1	1	1	1	1	1	1	1	1	1	2
Walsall	3	3	3	3	2	2	2	2	1	2	1
West Bromwich Albion	2	2	1	1	1	1	1	1	1	1	1
Wigan Athletic	2	2	3	3	3	3	2	2	2	2	2
Wolverhampton Wanderers	1	1	1	1	1	1	1	1	1	1	1
Wrexham	3	3	2	2	2	2	2	2	2	2	2
Wycombe Wanderers	[a]	[a]	3	2	2	2	2	2	2	2	2
York City	3	3	2	2	2	2	2	2	3	3	3

NOTE: 1 = Division 1, 2 = Division 2, and 3 = Division 3. Since the 1992-1993 season, the professional soccer leagues in England have been named the Premier League and Divisions 1, 2, and 3. During the 1992 season and before, the same leagues were named Divisions 1, 2, 3, and 4. To avoid confusion, the new league names have been applied to 1992.
a. Not a member of the Nationwide Football League.

The implication of the preceding observation is that turnover within the three divisions is extensive. Except for Premier League members, teams cannot conveniently be divided into a large majority that have a stable league assignment and a minority that are on the margins between two leagues and so alternate back and forth. The dramatic rise and fall of about one fourth of the teams in a decade, and the slower but real turnover that takes place among the teams that do not stay in the Premier League for long runs, show that during the course of a decade the promotion and relegation system has a substantial effect on the composition of all of the leagues. A surprisingly large fraction of teams in the lower leagues occasionally have runs that last several years in which they ascend to Division 1, and maybe even the Premier League for a few years, before falling back to Divisions 2 and 3. The next section examines recent history to determine the financial consequences of these runs.

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TABLE 5: Revenue Rankings of Promoted Teams: 1997-1998 and 1998-1999

	1997-1998		*1998-1999*	
League	*Promoted Team*	*Revenue Rank*	*Promoted Team*	*Revenue Rank*
Division 1	Bradford City	6	Charlton Athletic	14
	Sunderland	1	Middlesbrough	2
	Watford	10	Nottingham Forrest	5
Division 2	Fulham	2	Bristol City	2
	Manchester City	1	Grimsby Town	9
	Walsall	6	Watford	4
Division 3	Brentford	17	Colchester City	19
	Cambridge United	4	Leicester City	7
	Cardiff City	2	Macclesfield Town	NA[a]
	Scunthorpe United	8	Notts County	3

NOTE: NA = not applicable.
a. Macclesfield Town did not make its financial reports public in 1998-1999; however, since entering Division 3 in 1997, it has not been among the leaders in revenues in years when its revenues were reported.

INCENTIVES UNDER PROMOTION/RELEGATION

The preceding section documents the extensive turnover in league memberships, especially for Divisions 1, 2, and 3. Recall that the economic theory of a sports league points to an inefficiency of turnover: An assignment of teams among the leagues may not be a good match with relative local market demands. The promotions of 1997-1998 and 1998-1999 illustrate the problem. Table 5 shows that many of the teams that were promoted in these years were not among their league's leaders in revenues. Had promotion been based on revenues rather than on-field performance, fewer than half of these teams would have been promoted.

Because the allocation of teams among leagues does not maximize total revenues, the financial performance of the leagues exhibits several interesting patterns. As shown in Table 6, each league shows very large revenue disparities among teams in the league. In the Premier League, Manchester United has far more revenues than any other team, so revenues for the next best team are also shown. In all leagues, revenue disparities are larger than for major leagues in the United States.

As shown in Table 6, in every league a few teams have lower revenues than the revenues earned by the strongest teams in lower leagues. Indeed, in 2 of the 5 years, the revenue of the best team in Division 2 either surpassed or came within £1 million of the worst Premier League team, and in every year the best teams in Division 3 had roughly the same revenue as the worst team in Division 1. Table 7 shows the details of this relationship for the Premier League and Division 1. During the past 5 years, on average 3 Division 1 teams have captured more revenues than the weakest

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TABLE 6: Revenue Inequality in English Soccer: 1994-1995 to 1998-1999 (in thousands of pounds)

	1994-1995	*1995-1996*	*1996-1997*	*1997-1998*	*1998-1999*
Premier League					
Highest revenue	60,622	53,316	87,939	87,875	110,920
Next best	24,723	28,970	41,134	49,177	59,092
Lowest revenue	5,189	5,874	9,238	12,520	13,448
Ratio from highest to lowest	11.68	9.08	9.52	7.02	8.25
Ratio from next best to lowest	3.99	4.93	4.45	3.93	4.39
Division 1					
Highest revenue	7,565	10,450	12,727	18,825	24,078
Lowest revenue	1,697	2,173	2,152	2,111	2,271
Ratio	4.46	4.81	5.91	8.92	10.60
Division 2					
Highest revenue	6,942	4,441	5,061	4,533	12,731
Lowest revenue	827	941	799	1,127	1,243
Ratio	8.39	4.72	6.33	4.02	10.24
Division 3					
Highest revenue	1,860	2,876	2,847	1,886	2,043
Lowest revenue	541	528	722	717	702
Ratio	3.44	5.45	3.94	2.63	2.91

TABLE 7: Revenue Overlap Between Premier League and Division 1: 1997-1998 and 1998-1999 (in thousands of pounds)

			Number of Teams in Gap	
Year	*Highest Revenue in Division 1*	*Lowest Revenue in Premier League*	*Division 1*	*Premier League*
1994-1995	7,565	5,189	3	4
1995-1996	10,450	5,874	5	6
1996-1997	12,727	9,238	2	4
1997-1998	18,825	12,520	4	5
1998-1999	24,078	13,448	1	9

team in the Premier League, and on average 5.6 Premier League teams have had lower revenues than the Division 1 team with the highest revenues. In these cases, teams in the lower division draw better than other teams that finish ahead of them in the standings in the same division in 1 year and in the next year when these teams are promoted.

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The substantial overlap in team revenues among the leagues reflects two factors. First, in every year a fairly large number of teams play in lower leagues than their market demand justifies. Second, the revenue data are consistent with the hypothesis that the promotion/relegation system has a positive benefit for at least some teams in that revenues are enhanced by the prospect of promotion. A more formal econometric model is necessary to disentangle these effects completely but quite a bit can be discerned from the raw data.

The financial data provide clear evidence that league assignment affects revenues and so creates a powerful incentive to seek promotion. Table 8 contains revenues, attendance, wages, and net transfer fees for teams in years around their promotion and/or relegation between the Premier League and Division 1. The table segregates the teams into four groups according to how many times they were promoted and relegated. In every case both revenues and attendance increase when a team is promoted, with the average increase nearly £5 million. Attendance at home league matches is probably a better indicator of the revenue effect of league status than is revenues, for the latter reflects many other revenue sources, such as participation in tournaments and international matches and the "parachute" policy of giving some of the television revenues of the Premier League to teams that have been demoted in the previous 2 seasons. Hence, revenues can vary from year to year for extraneous reasons. On average, a promotion to the Premier League was accompanied by an increase in home attendance of about 6,000 per game. Of course, revenues and attendance in both leagues were growing during this period but not by nearly this amount.

Interestingly, demotion is not always accompanied by a drop in revenues and attendance. For the 5 teams that were permanently demoted from the Premier League, revenues increased slightly for 3 and fell by a large amount for 2, causing an average net revenue loss for all 5 of more than £400,000 per team. For teams that spent 1 year in the Premier League, revenues declined for 2 out of 3 and for the group as a whole by about £500,000 per team. For the three demotions among the 2 teams that changed leagues four or more times in consecutive years, two were associated with revenue declines and the average net loss was about £700,000. In all cases but one, demotion was accompanied by a drop in attendance, typically by a few thousand per game. Thus, demotion usually causes teams to be worse off financially; however, the drop in revenues and attendance is less than the gains from promotion, so that teams end up better off in the demotion year than they would have been had they not played in the Premier League in the previous season. These facts confirm that promotion has a hysteresis effect: The benefits of promotion persist after a team is demoted and average more than £4 million in the first year after relegation.

The meaning of the data on transfer fees and wages is not as clear as it is for revenues and attendance. Wages include all employees, not just players, and in particular the salaries of management. Hence, movements in wage data do not necessarily reveal the trend in total player salaries. Transfer fees reveal the attempt of teams to

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TABLE 8: Effects of Promotion and Relegation in Premier League, 1992-1999 (in thousands of pounds)

	Teams That Were Promoted Once and Stayed Through 1999 (wages and attendance unavailable)	
	Blackburn Rovers (1992)	*Newcastle United (1993)*
Revenues[a]		
Before	2,067	8,743
After	6,305	17,004
Net transfer fees[b]		
Before	(6,840)	(6,248)
After	(395)	1,366

	Teams That Were Demoted Once and Stayed Through 1999[c]				
	Manchester City (1996)	*Notts County (1992)*	*Oldham Athletic (1994)*	*Queens Park Rangers (1996)*	*Sheffield United (1994)*
Revenues					
Before	12,698	2,764	5,764	7,173	5,431
After	12,727	2,906	4,023	7,497	4,325
Net transfer fees					
Before	(2,588)	428	9	700	(46)
After	(1,754)	1,477	3,479	(3,556)	2,404
Wages					
Before	6,426	NA	3,088	5,010	3,146
After	7,200	NA	2,739	6,659	3,145
Attendance[d]					
Before	27,869	NA	12,563	15,683	19,462
After	26,753	NA	8,444	12,544	14,462

	Teams That Moved Twice: Promoted Then Relegated in Consecutive Years		
	Barnsley (1997-1999)	*Sunderland (1996-1998)*	*Swindon Town (1993-1995)*
Revenues			
Year 1	3,658	7,166	3,373
Year 2	12,694	13,415	5,415
Year 3	7,640	18,825	4,772
Net transfer fees			
Year 1	273	(1,812)	455
Year 2	(5,656)	(842)	(660)
Year 3	(512)	(2,018)	1,834
Wages			
Year 1	2,613	4,451	NA
Year 2	4,172	5,703	2,711
Year 3	7,077	8,161	3,393

(continued)

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TABLE 8: Continued

	Teams That Moved Twice: Promoted Then Relegated in Consecutive Years		
	Barnsley (1997-1999)	*Sunderland (1996-1998)*	*Swindon Town (1993-1995)*
Attendance			
Year 1	11,356	17,482	10,715
Year 2	18,449	20,865	15,274
Year 3	16,269	33,492	9,744

	Teams That Changed Leagues in 4 or 5 Consecutive Years				
	Year 1 (P)	*Year 2 (P)*	*Year 3 (R)*	*Year 4 (P)*	*Year 5 (R)*
Leicester City (1993-1997)					
Revenues	6,233	9,697	9,272	17,320	—
Transfer fees	(2,267)	1,418	(980)	(4,709)	—
Wages	3,132	4,421	5,449	8,914	—
Attendance	16,005	19,572	16,530	20,184	
Bolton Wanderers (1994-1999)					
Revenues	5,488	6,742	7,653	15,711	12,604
Transfer fees	(851)	553	(1,587)	623	1,323
Wages	2,682	3,312	6,159	9,625	10,039
Attendance	13,029	18,822	15,826	24,352	18,240

NOTE: NA = not applicable, P = promotion, and R = relegation.
a. Revenues are from all sources except sales of player rights (transfer fees).
b. Transfer fees are recorded as the difference between sales and purchases, and numbers in parentheses represent net purchases.
c. Norwich City is also in this category but insufficient data are available to include it.
d. Attendance is average per league match.

improve their team (a positive number) or to sell quality players (a negative number); however, these numbers also are affected by extraneous events, such as injuries, retirements, and unevenness through the years in the number of players who seek employment with another team after their contracts expire. In some cases, teams seriously seek to improve themselves to stay in or return to the Premier League, as reflected in large expenditures on player transfers and higher wage bills.

Notwithstanding the caveats, the data indicate that teams do not all pursue the same strategies in response to promotion and relegation. In some cases, teams take advantage of their promotion to sell the players that made promotion possible and consequently stay briefly in the Premier League. In other cases, teams on the cusp between Division 1 and the Premier League put forth great effort to gain and keep promotion.

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The Bolton Wanderers exemplify the first strategy. In the years before their two promotions in the 1990s, they had substantial net expenditures on transfer fees and then had net sales of players during the years they were in the Premier League. They also were net sellers of players when they were demoted in 1999.

The Wolverhampton Wanderers have a similar history to Bolton but with a different outcome. Wolverhampton is 1 of 3 teams that played in Division 1 throughout the 1990s and is always among the leaders in attendance. In the past 5 years, Wolverhampton's rank in revenues among Division 1 teams has averaged 2.4. Twice it has had the highest ranking, and in all years its revenues have surpassed the revenues of the lowest ranking Premier League teams. Nevertheless, Wolverhampton has not been promoted to the Premier League, finishing 3rd, 20th, 4th, 7th, 9th, and 12th (average 8.9) during the past 5 years.[10] Wolverhampton's stable membership in Division 1 has three possible explanations: luck (the difference between Wolverhampton and Bolton, with similar team quality, may be due to randomness in which teams are promoted), management that does a slightly better job of calibrating team quality (if promotion, on balance, is unprofitable), or management that does a slightly worse job of controlling team quality (if occasional promotion is profitable).

Manchester City apparently pursued the opposite strategy of Bolton and Wolverhampton. The team was a large net purchaser of player contracts in both its final year in the Premier League and its 1st year in Division 1, but to no avail as it slipped all the way to Division 2 two years after its demotion; however, the team did climb back to the Premier League for the 2000-2001 season but was then relegated in 2001. Likewise, Sunderland tried first to stay in and then to return to the Premier League by being a substantial net purchaser of player contracts and eventually succeeded for the 1999-2000 season, climbing to finish 7th in the Premier League in 2000-2001.

Leicester City seems to have pursued both strategies. In 1993-1994, the team spent heavily on net transfer fees and improved enough to ascend to the Premier League. But in 1994-1995, the team had a net gain of £1.4 million in player sales (more than enough to offset the higher salary bill it paid that year) and was demoted back to Division 1. After demotion, the team once again spent a substantial amount on transfer fees and was promoted. But the second time the team spent still more on salaries and transfer fees and succeeded in sticking in the Premier League, where it remains 6 years later.

The propensity of some teams to weaken themselves after promotion is to be expected from the fact that some promoted teams are not among the revenue leaders of the league from which they were promoted. The yo-yo teams plausibly operate in markets that cannot sustain competitive teams in the Premier League, so when they are promoted they do not make a serious attempt to stay. In this case, the motivation for promotion is not to obtain a permanent place in the higher league but to take advantage of the hysteresis effect in attendance. For teams in lower leagues that have higher revenues than some teams in higher leagues, the incentives are likely to

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be different. Their best assignment is likely to be in the higher league, and so they are more likely to make the expenditures that are necessary to increase their chance of staying in the higher league. The puzzling case is Wolverhampton, which always does better financially than several Premier League teams yet seems to have found a permanent home in Division 1.

IMPLICATIONS

The financial and playing field experiences of the four leading British football leagues are generally consistent with the predictions of the theoretical model that can be tested using the financial data that are available. Promotion/relegation clearly has led to turnover in league membership among the four leading leagues; however, promotion and relegation between Division 3 and the National Conference has led to very little successful entry into the top echelon of teams. Apparently the combined size of the four leading leagues is sufficiently large that almost all viable franchise locations—especially for the top two leagues—were occupied early in the 20th century. Nevertheless, turnover between the Premier League and lower leagues is substantial in the time frame of a decade. Teams that play in rapidly growing markets or that switch from poor to strong management can rapidly ascend from Division 3 to the top leagues.

The financial information indicates that promotion is financially attractive as well as desired by fans. Both revenues and attendance at league matches tend to increase substantially when teams are promoted. Moreover, the reward from promotion seems to endure for a while after a team is demoted, giving teams that are marginal for a higher league a financial incentive to field teams that bounce back and forth between a higher and lower league.

Of course, this article is only a beginning. The next step is to estimate econometric models of team performance, revenues, attendance, wages, transfer fees, and other financial variables to provide more precise estimates of the effects described above and to determine whether the noisy indicators of player costs can sustain tests of the effects of promotion and relegation on team strategies regarding players.

Applications to North America

Drawing conclusions about United States and Canadian sports from the experience of English football is hazardous and, as a practical matter, probably irrelevant because North American leagues are not likely to experiment with so dramatic a change in format as the introduction of promotion and relegation. Nevertheless, the English football system seems to be a very attractive solution to the perpetual problem of leagues with substantial disparities in on-field success and revenue among their teams.

Promotion and relegation have three significant advantages and one significant disadvantage. On the plus side, this system allows the top league to be smaller, have

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higher average revenues, and be more balanced competitively, although the latter effect is muted by the yo-yo teams that seek only occasional, brief promotion to profit from the hysteresis effect on revenues. On the negative side, teams that are relegated do worse financially than they would if they finished in the bottom group of teams in the higher league. This disadvantage may be offset by the advantages but the issue cannot be resolved without running an experiment.

A promotion/relegation system can be implemented in two ways: by constructing a lower league from the existing minor league system or by dividing the existing major league. Sports with independent minor league teams are the most likely candidates for the first method.

Changing relations with minor leagues is not likely to be viewed favorably in baseball. The vast majority of minor league baseball teams, and all in the top two classifications, operate as farm teams for major league franchises. This structure enables major league teams to monopsonize the market for players who have any serious chance of becoming major leaguers through the amateur draft. Farm teams are expensive—MLB teams spend around $12 million per year on scouting amateurs and providing players to minor league affiliates—but this cost is more than offset by the salary suppression that arises from the fact that players must have 3 years of major league experience before they qualify for salary arbitration and 6 years before they become unrestricted free agents. In addition, baseball has some attractive open markets that are valuable as potential expansion sites, most notably Northern Virginia and Northern New Jersey, which would be lost if minor league teams in those areas could qualify for promotion to the majors. Thus, baseball does not appear to be a good candidate for rearranging its minor league structure to accommodate promotion and relegation.

Football has three significant minor leagues: Arena Football (indoors), the Canadian Football League (CFL), and the World League of American Football (in Europe). The NFL owns Arena Football and the World League, but the CFL is independent. Because of the difference in format, Arena Football is inappropriate for a promotion/relegation system with the NFL; however, both the CFL and World League are good candidates.

The advantages to the NFL of adopting a promotion/relegation system are that it provides a mechanism for expanding into Canadian and European markets that are already proven and it enables the NFL to sell World League franchises at a substantial premium. Likewise, the CFL, which tried unsuccessfully to enter the United States in the mid-1990s, probably would jump at the chance to become a Division 1 to the NFL's Premier status. The NFL almost certainly would insist that neither the CFL nor an independent World League place teams in the territories of existing NFL franchises, and in return would promise not to expand or to relocate teams to the areas occupied by teams in these leagues.

The disadvantage of this plan to the NFL is that it would lose the high expansion fees that await it in U.S. cities that are likely to be a viable home for an NFL team. Most notably, Los Angeles almost certainly would become an expansion target of

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the CFL if teams in that league could be promoted to the NFL. Whereas the NFL has few open markets that are as attractive as Los Angeles, the experiences of the Jacksonville and Carolina expansion teams and the relocation of the Houston Oilers to Nashville indicate that NFL teams can be financially viable in a large number of cities. As a result, the NFL has more plausible expansion cities remaining in the United States than any other sport; however, the loss of these expansion fees, due in part to the fact that expansion cannot be much faster than two teams every several years, would be partially offset by the enhanced value of World League franchises.

Both basketball and hockey provide interesting (if still not very plausible) possibilities. Both sports have independent minor leagues, but in both sports these leagues have disadvantages as candidates for promotion and relegation. Farm relationships with teams and the major leagues are common in hockey. Basketball's best minor league, the Continental Basketball Association, failed during the 2000-2001 season. Favoring a promotion and relegation system are the fact that basketball and hockey have expanded about as far as is plausible and have several franchises that play in weak markets and/or field persistently weak teams, so they have little to lose by experimenting with promotion/relegation.

The second approach to implementing a promotion/relegation system is to divide the existing major leagues to create a significantly smaller top league. The basic idea would be to shrink the top "major-major" league to something between 16 and 20 teams. Perhaps in tandem with expansion or promotion of some minor league teams, the remaining major league teams would be placed in a "minor-major" league with a comparable number of teams. All teams in both leagues would be treated as a "major league" in that all would participate in league governance, would have the same relationship with minor leagues, and would participate in the market for major league players. As the Premier League did when it shrunk to 20 teams and Division 2 did when it reorganized from two coequal leagues to the present Divisions 2 and 3, leagues would implement this system by demoting the major league teams with the weakest records in the season before the plan was implemented. To make the system more palatable to owners, players, and fans of demoted teams, the system could promote the 4 best teams in the lower league at the end of each season.

In 2001, baseball announced a plan to disband at least 2 and maybe more franchises as its solution to revenue disparities in the sport. This action seems to have been motivated primarily by the desire to stop sharing local revenues with low-revenue teams. Moving to a hierarchy of leagues with promotion and relegation achieves the same objective without eliminating teams. A structure of two 8-team leagues at both the major-major and minor-major level, each with two divisions, would enable baseball to return to a better balanced schedule: 16 games against each opponent in the same division, 12 games against each team in the other division of the same league, and 8 games against each team in the other league, for a

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total of 160 games. The League Championship series would be against the division winners of the higher leagues, followed by the World Series.

The loss of the first round of the playoffs could be replaced by a tournament resembling the FA cup in England, in which teams in both major-major and minor-major leagues play in a single-elimination tournament with matches based on a random draw rather than seeding. This tournament could be scheduled by creating a 1-week hiatus in the regular season schedule during the summer, before the All-Star game. To create an incentive to win, revenues from the broadcast rights to these games could be divided unequally so that teams and players receive a significant financial bonus for winning each game.

Basketball and hockey also might benefit from splitting into a two-league hierarchy. Both sports have several franchises that are financially weak and persistently doormats in the league standings, and that have few remaining opportunities for expansion or relocation. Thus, these sports also seem to have relatively little to lose by dividing into a hierarchy of two leagues and implementing promotion and relegation.

As in baseball, a smaller league would enable basketball to create a second championship, patterned after the National Collegiate Athletic Association (NCAA) basketball tournament (but without seeding) as a series of single-elimination games during a week in the middle of the season. Both the draw for matches and the location of the games (which would be the home team) would be determined randomly to give a small advantage to some weaker teams. As with baseball and NCAA basketball, the television proceeds for the tournament could be allocated on the basis of games won rather than shared equally, to create an incentive to take the tournament seriously.

Because the worst NFL teams remain financially strong and because the opportunity for a second type of championship is less attractive, shrinking the size of the major-major league is not attractive in football. The better model for the NFL probably is to create promotion/relegation deals with the two existing leagues.

In summary, promising avenues for implementing a system of promotion and relegation are available in all four sports. The main problem is that all require a substantial reorganization of the existing structure of the major league. Reforms of such major proportions, no matter how attractive, are not likely to be implemented because doing so would require supermajority support from the existing set of major league teams. In all sports, finding any reform that commands overwhelming support is very difficult, so there is little immediate prospect that these reforms would even be considered, let alone adopted. Nevertheless, contemplating their likely effects is an interesting and instructive exercise, particularly in light of the hand-wringing by owners in each sport concerning their financial plight.

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APPENDIX

The data reported in this article come from several sources.

The playing histories for the Nationwide Football League are from a Web site that contains the historical records of all of the major European leagues, www.soccer-links.com/info/. The histories of the Nationwide Football League teams, as well as standings for all English leagues (including those below the top four), can be found in Glenda Rollin and Jack Rollin's *Rothmans Football Yearbook 2000-2001* (1969/2000). Another useful source of information is the Rec.Sport.Soccer Statistics Foundation, especially the data compilations on historical performance information by Ross (1996) and Pietarinen (2001).

Financial information comes from an annual compilation of the financial statements of nearly all British football clubs that is edited by Gerry Boon, the *Annual Review of Football Finance* (1996/2000). English law requires financial disclosure for all corporations, and the publisher Deloitte and Touche compile these data annually. The major problems with the financial data are that some teams provide too little information to permit meaningful comparisons of some elements of all financial reports and that some teams release their reports too late for inclusion in the volume. About a dozen of the 92 teams fall into one of these categories each year. Deloitte and Touche attempts to reconcile reports that use different accounting systems, but these attempts are limited by data availability. The series on wages is of dubious value because it lumps together players, coaches, and all other personnel. Likewise, the data on transfer fees were of limited usefulness until recently, when the British government imposed a uniform system of accounts on sports teams. As is the case for sports teams in the United States, the data on profits were not very useful because of differences among the teams in practices regarding administrative costs and related businesses.

NOTES

1. Secondary criteria are a team's stadium quality, revenue potential, and adherence to league rules. In some cases relegated teams are not those with the worst records, and teams are denied promotion because of operational shortcomings.

2. New U.S. leagues that managed to have at least one team enter an established major league were the American League and the Federal League in baseball, the Basketball Association of America, the first American Basketball League (1947-1949) and the American Basketball Association in basketball, the All-American Football Conference and the American Football League in football, and the World Hockey Association in hockey. Only one entrant, the Basketball Association of America (BAA), has managed to supplant the incumbent as the dominant league. In the late 1940s, the BAA merged with the incumbent National Basketball League and the first American Basketball League to form the National Basketball Association. Most of the teams in the merged entity were from the BAA. See Noll (1991) for a history of this era.

3. Entry into the lowest league requires approval, but low leagues do not charge high entry fees.

4. The underlying assumption is that teams are price-takers in inputs to team quality, which implies that each team operates in a competitive market for players, coaches, executives, and stadiums. The model developed by Ross and Szymanski (2000) assumes that leagues also face a perfectly elastic supply of quality so that all of the effects of promotion and relegation on demand cause increased quality rather than higher wages. The rapid escalation in soccer player salaries during the past few years as reve-

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nues of top teams have grown suggests that, in fact, much of the increased demand for quality leads to wage increases.

5. Although industry leaders and some scholars dispute the profit-maximization assumption, this debate is not important for our purposes. Because many soccer teams are organized as publicly traded corporations, the profit-maximization assumption seems unexceptional. But even if owners have more complex motives, these can be captured by the standard model by assuming that the revenue function, $R(q)$, has two components: funds received, F, and the willingness to pay of ownership to have a victorious team, V. If both F and V are functions of team success, and expected success is a function of team quality, the profit-maximization problem remains maximizing $R(q) - wq = F(q) + V(q) - wq$, and the conclusions in the main text remain valid.

6. If Team 1 enjoys more intense demand than Team 2 and if the teams have competitive balance (i.e., if $q_1 = q_2$), then $R'_1 > R'_1$ so that both cannot satisfy Equation 2.

7. This result is the same as the result in Ross and Szymanski (2000). The latter assumes an explicit form for the revenue function and derives the magnitude of the incentive effect on team quality under promotion and relegation.

8. For example, to obtain an antitrust exemption for their merger, the National Football League and American Football League granted an expansion franchise to New Orleans, the home of Senator Russell Long, who sponsored the legislation.

9. The history of the team cheerily is told on its Web site, www.thediamonds.com.

10. In 1994-1995 and 1996-1997, Wolverhampton qualified for the playoffs for promotion but lost.

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